



Unit 4: Risk Management

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Introduction to the Unit

In this unit, we will examine the *Risk Management Process* and its various components. Upon completion of this unit, we will have a fair understanding of the various methods of identifying risks and the possible strategies for mitigating them.

We will begin this unit by bringing out the differences between *Constraints* and *Risks*, followed by an introduction to the role of *Risk Management* in the overall *Project Management Life Cycle* and will complete the learning cycle getting an in-depth understanding of the various steps involved in risk *Identification, Analysis and Mitigation*.

Objectives

Upon completion of this unit, you will be able to:

- Differentiate between constraints and risks
- Describe the importance of the Risk Management Process
- Identify the components of the risk management plan
- Identify the steps involved in identifying risks
- Explain the process of analysing risks
- Identify the strategies for mitigating risks



Constraints and Risks

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Objectives

Upon completion of this segment, you will be able to:

- Describe what constraints are
- Describe what risks are
- Explain how risks are categorised

Introduction

In the last segment, we learned about the importance of measures and metrics and how they are used to measure the project performance. In this segment, we will briefly understand what is meant by constraints and risks in the context of project management.

From initiation to the closure of a project, constraints and risks set the stage for project planning and execution. Both of these are ground realities that a *Project Manager* must keep in mind and, accordingly, make appropriate decisions to keep the project in good stead.

Let's, briefly, understand both.

Constraints

A constraint is a limiting condition, circumstance or event that sets boundaries on the project process as well as on the expected results.

Constraints are limitations on a project. Every project has a timeline (in which it has to deliver), scope of work (that needs to be completed) and cost (within which the project has to be completed and made successful). These three constitute the key constraints of a project.

In the project management lexicon, schedule, scope, cost and quality—which form the boundary of the project—are also its constraints. Once the project is kicked off and the *Project Manager* is provided all support to execute the project, he/she has to monitor these closely.

Along with scope, schedule and cost constraints, the *Project Manager* has to manage other constraints related to resources, risks and issues.

In the case of a wedding ceremony, what could be the constraints faced by a wedding planner?

Finding good resources such as an experience caterer and a priest could be some of the many constraints that a wedding planner is likely to face.



Could rain be a constraint for the marriage?

It could be—depending on whether the ceremonies are arranged indoors or outdoors.

We will discuss more about constraints—and how to manage them—in later segments; however, our key takeaway—so far—from this segment should be that constraints are one of the ground realities of project management and the challenge for the *Project Manager* is to work against all constraints and make the project a success.

Risks

Risk is another term that is commonly used in project management as well as in everyday life. One of the widely available definitions is that ‘risk is a probability or threat of damage, injury, liability or loss caused by external or internal forces, which requires some action.’

Can this always be true?

If you were to throw an open-air party for your friends, in the rainy season, would you not be taking a risk? Well, if you and your friends had a rain dance party on your minds then it is definitely not a risk. But, for many others, rain could ruin the party.

You assume that it is not going to rain till the end of the party and go ahead. However, while you are risking the possibility of rain, you can either plan to carry umbrellas or stay close to some covered area.

Therefore, while there is an assumption and a risk, there is also a plan to manage the risk so that the activity is not interrupted.

Most of the assumptions are taken forward and evaluated as risks. For every risk, there can be a plan to either mitigate it before it occurs or to have a contingency plan if it has already occurred.

This is the core concept of risk management!

The following is a list of some typical risks:

- The *Project Manager* of a big project has—so far—managed only small projects.
- The organisation has not proven that it has the right skill set for a project.
- The employees may not have received the necessary training.
- There could be a major strikes during the period of work.
- Budget cuts might happen for the next fiscal year.
- End user may insist on new requirements at the last moment.
- Additional requirements may need to be added midway in a project.
- Requirements are felt to be more complex than perceived earlier.
- There is attrition in the team, midway, during a project.
- Machinery may not arrive on time
- Legal regulations might change during the project lifespan.
- Work place environment may be unsafe.
- Work is begun before all stakeholders have agreed and signed off.

Projects have to sail through many risks to be successful. During project initiation, project sponsors evaluate the risk of undertaking a project. However, all risks are not identified during the initial



phase of the project. There will be many unknown factors in the initial phase which will be documented as assumptions and risks. The *Project Manager* assigns risk owners for them and regularly follows up. He/she creates an efficient plan to manage risks such that even before they come true, he/she has an action plan that can either reduce its impact or accept it with minimal loss.

In the project life cycle, the number of risks in the initial period will be high. As the project progresses and phases, the numbers reduce. However, impact and priority of risks can keep varying. It is not true that only the risks that are identified during the early stage have to be managed. The risks that are foreseen during other phases of work also have to be recognised and managed effectively.

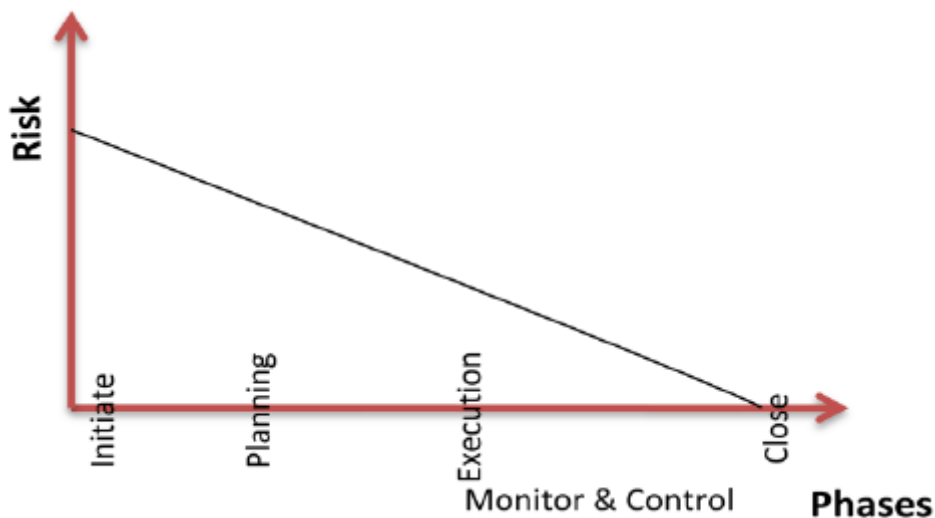


Figure 28: Risk—Through the Stages

Are all risks negative?

Not all identified risks have a negative impact on the project. Some of the risks can also have a positive impact. Certain risks can enhance business, bring in more customers and revenue, generate goodwill among users and so on.

Accordingly, the definition of risk also changes to: ‘risk is the likelihood of any event which has either a positive or a negative influence on the project.’

Positive risk is almost a philosophical idea for many. But it is really how you look at it. There are people who believe that there is no such thing as a positive risk. Many others maintain that positive risk is anything that offers an opportunity.

Just look at this example and think about what the risk is and how it getting converted into a positive risk.

A nightclub away from the major cities is only frequented on Fridays and Saturdays. They start a promotional campaign to introduce Thursday nights by cutting the prices dramatically.

The promotion team manages the risk that Thursday nights will not be successful. They also manage the positive risk that Thursdays will be so successful that customers will stop showing up on the weekend.



Project Management

Positive risks can take a number of forms but they can be hard to uncover as our brains are conditioned to think of risks as bad. The easiest way to identify a positive risk is to do it the same way as we would identify a negative risk.

We must analyse a risk properly and come up with a list of opportunities that could impact the project. The next stage is to brainstorm about all the good things that could happen.

The following events are examples of what can also be interpreted as positive risks:

- Receiving so many calls about a new product that increase in sales can be anticipated.
- Getting 10 times more hits on a website—than was originally anticipated—due to excellent publicity.

Risks can be classified in the following categories:

- **Negative and positive risks:** Based upon their impact.
- **Internal and external risks:** Based on the source of the risk.
- **Catastrophic, critical, marginal and negligible risks:** Based on the magnitude of the impact.

A software project may classify its risks as:

- technical,
- functional,
- operational,
- financial, and
- legal.

Similarly, a construction project may classify some of its risks as:

- supply of material,
- vendor-related risks,
- labour-related risks,
- financial risks, and
- safety-related risks.

Organizations may classify risks based on various factors in keeping with their business and knowledge.

Figure 29 shows a generic classification of risks.

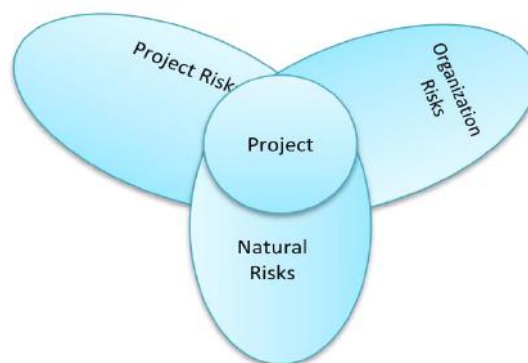


Figure 29: Classification of Risks



Project Management

Project risks involve risks due to improper requirements, resources, cost and quality. Organisational risks are related to managerial decisions, work hours and support risks that need to be provided by the organisation.

Natural risks are related to natural calamities such as earth quakes, weather changes and any other risks caused by natural forces. These could affect the project either through the resources or in any other form.

The *Project Manager* with the help of SMEs—in those areas—has to identify risks and log them in the risk register. It is important that risks are classified and their possible impact on the project is documented.

Based on the impact and priority, an appropriate response needs to be planned. This needs to be followed by a detailed risk plan that should help the *Project Manager* to manage risks appropriately during the course of project execution.

A mitigation plan is a risk response plan to mitigate in such a way that the possibility of risk is averted. A contingency plan is a risk response plan which necessitates immediate action after a risk materializes, so that its impact can be reduced.

We will discuss more about risk plans in the segment on Risk Management.

Summary

In this segment, you learned that:

- a constraint is a limiting condition, circumstance or event that sets boundaries on the project process as well as on the expected results;
- risk is a probability or threat of damage, injury, liability or loss caused by external or internal forces, which requires some action;
- the number of risks decrease as a project moves from the initial stage to the later stages;
- risks can be, both, negative and positive; and
- various organisations categorise risks differently, depending upon the kind of challenges faced.



Risk Management

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Objectives

Upon completion of this segment, you will be able to:

- Explain the role of risk management in overall project management
- Identify the steps necessary in risk management
- Describe the process of risk identification and analysis
- Identify the key strategies for mitigating risk

Introduction

As we have understood in the previous segment, risk management is a field of management that deals with the possibility that various future events may cause harm or threat to the organisation. Risk management comprises strategies and techniques to recognise and confront any threat faced by a business in fulfilling its mission.

Risk management deals with following three basic questions:

- What might go wrong?
- What can we do (both preventive and corrective measures to deal with the incident must be thought of)?
- If something unexpected happens that upsets our plans, how do we tone down its effect?

Risk management is concerned with the effective management of risk data and information of an organisation.

Risk management information systems/services (RMIS) are often used by enterprises to provide expert advice and cost-effective information management solutions. It deals with key processes such as:

- risk identification and assessment,
- risk quantification, and
- risk control.

These points are explained in greater detail in the later part of this segment.



Risk management plans

An effective risk management plan comprises the following steps:

Creation: To create a risk management plan, we must carefully select appropriate controls and counter measures to quantify each risk. Risk mitigation must also be approved by the suitable level of management as per the level/domain of risk.

For example, let us suppose that there is a risk that is likely to harm the reputation of an organisation; it would be best dealt with by the top management. However, if there is a risk to the computer system or the information system because of a computer virus then the IT department would prove to be the most competent and will have the authority to deal with it.

The risk management plan must also suggest valid and effective security controls for managing the risks. For example, if there is a high risk of computer viruses, it could be mitigated by installing an antivirus software on the system.

A good quality risk management plan consists of a schedule for the control operation and people accountable for those actions.

As per ISO/IEC 27001, after the commencement of the risk assessment phase, a risk treatment plan must be designed. A risk treatment plan must clearly state the decisions about how each of the identified risks are to be dealt with.

Implementation: The next step is the implementation of the plan. Firstly, we must undertake measures for mitigating the effect of the risks. We must purchase insurance policies for all those risks that could be insured. You must try to avoid and minimise all risks without sacrificing the entity's goals. The risks left after all these, are retained.

Review and evaluation of the plan: The initial risk management plans are certainly not going to be perfect. The actual risk control measures are largely based on a number of factors and, hence, there is always a high probability that the management might have to consider changing its plan.

Also, it is necessary that the risk results and management plans are periodically updated. It is basically done for two underlying reasons, namely:

- to find out if the previously selected security measures are still valid and effective, and
- to determine the possible risk level changes in the business atmosphere.

For example, market risks are an example of the rapidly changing business environment.

The Role of Risk Management in Overall Project Management

Risk analysis and management is a process which enables analysis and management of the risks associated with a project. Properly undertaken, it will increase the likelihood of successful completion of a project in keeping with the cost, time and performance objectives.

There are a lot of benefits of proper risk management in projects. Organisations can generate a lot of profit if they deal proactively with uncertain project events. We can deliver a project on time, within the pre-defined budget and with proper quality if we are able to manage the risks properly.



Proper risk management can increase the productivity and efficiency of the *Project Team*.

As we have learned so far, a project life cycle includes key phases such as initiating, planning, executing, controlling and closing. The probability of project risk depends on the project life cycle.

Steps in Risk Management

Effective risk management involves the following key steps:

Step 1 – Recognition of the assets at risk: The foremost step in the risk management technique is to carefully identify the assets which might generate risks in project operations. These assets may fall under various groups, such as tangible and intangible assets, movable and immovable assets and so on.

Step 2 – Valuation of assets: The assets identified and grouped in the previous step are to be valued and categorised into different classes such as critical and essential.

Step 3 – Identification of the intimidators: Threats can be distinct such as anything that contributes to the intermission or devastation of any service/product. Various compulsions can be grouped into environmental, internal and external threats.

Step 4 – Risk consideration: The process of risk appraisal includes not only assessment as to the provability of occurrence but also the assessment as to the impending severity of loss, if risk materialises. This will support in determining the appropriate risk lessening strategy, the residual risk and the investment required to alleviate the risk.

Step 5 – Application of emergent strategies for risk management: After risk identification and assessment, one must apply various risk management techniques such as risk avoidance, risk reduction, risk retention, risk transfer and so on.

Risk Identification

Risk identification is done at each stage of a project life cycle. During risk identification, we identify and categorise the risks. Risks must be carefully identified with the help of techniques such as brainstorming or reviewing a standard list of risks. Risk identification must be performed by the concerned people such as the IT personnel, marketing managers or the top-level management.

There are basically two types of risks. These include the following:

Business risks: These are the ongoing risks that are best handled by the business. For example, if a key team member becomes unavailable or sick then it may delay the project and the organisation might not be able to complete the project in the given financial year.

Generic risks: Generic risks are those that are common to all projects. For example, a system failure or a flaw may cause the project to be delayed.

Risks must be defined in two parts. The first part must define the cause of the risk and the second must define the impact of the risk. For example, a risk may be defined as 'The supplier not being able to meet a deadline will mean that the budget will exceed'. If the above format is used, then it would be much easier to remove duplicates and understand the risk.



For a comprehensive identification of risks, we may adopt the risk matrix suggested by Well-Stam et al. The vertical axis of this matrix represents the various phases of the project and the horizontal axis represents the various points of view or perspective. For an infrastructure project, the points of view may be technical, organisational, zoning, political, administrative, legal/legislative, financial/economic or social/community related.

For each class (phase and point of view), we may use some proven risk-identification techniques quoted in the literature for identifying the possible risks.

These techniques include:

Assumption analysis: Assumptions made in the planning stage of the project are taken as true, real or certain. A closer scrutiny of these may reveal possible risks.

Brainstorming: Brainstorming is a useful tool for quickly generating the possible risk events. It is performed by a cross-functional team by following certain set procedures.

Developing a checklist: The checklist is developed based on past experience. It provides a useful guide in listing the foreseeable risks.

Using a Delphi study: A Delphi study is carried out with the help of a group of experts. Since the experts are people with a deep insight into the functioning of the system, it is possible to gather useful information in this way.

Conducting interviews: An Interview may be held with knowledgeable people to identify or to gain more in-depth knowledge of certain risks or to create a list of control measures.

Making observations: We may visualise the risks by directly examining/observing the current process.

Referring to previous documentation: Past experiences recorded in company files, reports, third party reports, or newspaper reports in an electronic or a paper format provide help in listing the risks.

Modelling: The Use of risk tool kits or simulations by a computer or other aids may also be used for uncovering risks. There are a number of diagrams such as the cause and effect diagram, fault tree, event tree, influence diagram and so on, which can be used to capture risks.

Risk Analysis

A large number of risks may emerge during the risk identification process. It is not prudent to give equal attention to all the identified risks. Hence, there is a need to evaluate and prioritise them. The risk needs to be quantified in two dimensions, namely the impact and the probability of the occurrence of risks. For simplicity, we can rate each on a scale of 1 to 4. The larger the number, the bigger is the impact or probability. By using a matrix, a priority can be established.

Figure 30 depicts the risk analysis.



Probability	4	Medium	Critical		
	3				
	2	Low	High		
	1				
		1	2	3	4
		Impact			

Figure 30: Risk Analysis

It must be noted that when probability is high, and impact is low, it is a medium risk. Contrary to this, if impact is high, and probability low, it is high risk. Even the slightest chance of a disaster requires more attention than a high chance of a hiccup.

Reducing Risks

To make the most of the benefits of project risk management, we must integrate the project risk management activities into a well-planned project management plan and work activities. After the risks have been identified and measured, the strategies for controlling the risks can be broken down into one or more of the following categories:

Risk avoidance: It includes not performing an activity that could carry risk. For example, not buying a property or business to avoid the liability attached to it or not flying an airplane to avoid the risk of a crash.

Avoiding activities may seem to be a very easy way of dealing with risks, but it also means losing out on the potential gain that performing the risk-prone activities may have brought forth. For example, not entering a business to avoid the risk of loss also effectively ends the possibility of earning profits.

Risk reduction: It involves methods that reduce the severity of loss from occurring.

For example, sprinklers are designed to put out a fire to reduce the risk of loss by fire. This method may cause a greater loss by water damage and, therefore, may not be suitable. Halon fire suppression systems may mitigate that risk, but the cost may be prohibitive as a strategy.

Risk retention: It involves accepting the loss as and when it arises. True self insurance is an example. Risk retention is a feasible strategy for small risks where the cost of insuring against the risk is likely to be higher over time than the total losses sustained.



Project Management

Risks which are not avoided or transferred are retained by default. This comprises risks that are so disastrous that they either cannot be insured against or their premiums would not be feasible.

Also, any amount of potential loss (risk) over the amount insured is a retained risk. This can be accepted if there is a small chance of a very large loss or if the cost to insure for a higher coverage is so high that it would hamper the goals of the organisation.

Risk transfer: It means causing another party to accept the risk, usually by means of a contract or by hedging. An example of a risk that uses contracts is insurance. In other cases, it may involve contract language that transfers a risk to the other party without the payment of an insurance premium. Very often, the liability among construction or other contractors is transferred this way. On the other hand, taking offsetting positions in derivatives is normally how firms use hedging to manage a risk financially.

In this way, risk analysis and management proves to be a process, which enables the analysis and management of risks associated with a project. Properly undertaken, it can increase the likelihood of successful completion of a project in conformance with the stated cost, time and performance objectives.

Summary

In this segment, you learned that:

- risk analysis and management is a process, which enables the analysis and management of risks associated with a project;
- creation, implementation and review and evaluation are the key elements of an effective risk management plan;
- risk analysis is based on the quantification of risks according to impact and probability of their occurrence.
- avoidance, reduction, retention and transfer constitute the key strategies for risk mitigation.